

TrES-3b

R-0316

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_170523 · created 2026-07-18T00:17:51+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457896.7842 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.057
Transit depth (Rp/Rs)^2	3.87 +/- 2.22 [%]
Orbital Inclination (inc)	84.73 +/- 2.87
Airmass coefficient 1 (a1)	1.26 +/- 0.64
Airmass coefficient 2 (a2)	-0.19 +/- 0.22
Scatter in the residuals of the lightcurve fit is	49.89 %
Best Comparison Star	#3 - [384, 301]
Optimal Aperture	1.93
Optimal Annulus	3.75
Transit Duration (day)	0.071 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.197 $\pm$ 0.057 vs 0.16309 (deviation 0.55 σ)
Duration fit vs published	0.071 d vs 0.0567 d (deviation 0.73 σ)
Mid-time O-C	-3.9 min
Depth SNR	3.2
Residual scatter	49.89 %

Light curve

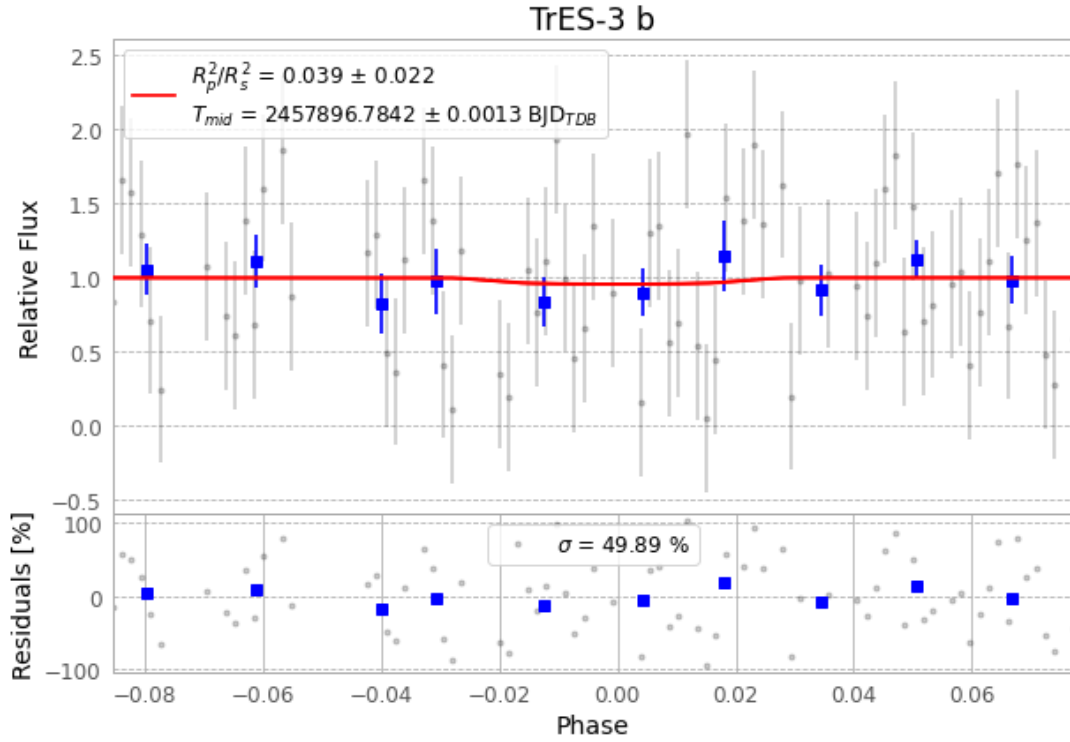


Figure 1 — FinalLightCurve_TrES-3 b_2017-05-22.png (SHA-256 c067b11cfe38568e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 192], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 47e2954b96c58176...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

103 FITS frames (68 MB), TRES-3170523040312.FITS → TRES-3170523090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-170523.log (2739b6d58d35...), FinalLightCurve_TrES-3 b_2017-05-22.png (c067b11cfe38...), FinalLightCurve_TrES-3 b_2017-05-22.csv (978fd7547b36...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 00:17Z.